

# DMFR SECURITY

another security blog...

## picoCTF 2019 vault-door-1 Writeup

ON NOVEMBER 1, 2021 / BY DANIEL / IN CTF, REVERSING  
vault-door-1 is a Reverse Engineering challenge worth 100 points.

Its description is:

This vault uses some complicated arrays! I hope you can make sense of it, special agent. The source code for this vault is here:  
VaultDoor1.java

A Java source file is provided, named `VaultDoor1.java`. This challenge was straightforward. Reading the source showed a silly comment and the scrambled password:

```

emacs27@wildcat
File Edit Options Buffers Tools Java Help
// I came up with a more secure way to check the password without putting
// the password itself in the source code. I think this is going to be
// UNHACKABLE!! I hope Dr. Evil agrees...
//
// -Minion #8728
public boolean checkPassword(String password) {
    return password.length() == 32 &&
        password.charAt(0) == 'd' &&
        password.charAt(29) == 'a' &&
        password.charAt(4) == 'r' &&
        password.charAt(2) == '5' &&
        password.charAt(23) == 'r' &&
        password.charAt(3) == 'c' &&
        password.charAt(17) == '4' &&
        password.charAt(1) == '3' &&
        password.charAt(7) == 'b' &&
        password.charAt(10) == 'l' &&
        password.charAt(5) == '4' &&
        password.charAt(9) == '3' &&
        password.charAt(11) == 't' &&
        password.charAt(15) == 'c' &&
        password.charAt(8) == 'l' &&
        password.charAt(12) == 'H' &&
        password.charAt(20) == 'c' &&
        password.charAt(14) == 'l' &&
        password.charAt(6) == 'm' &&
        password.charAt(24) == '5' &&
        password.charAt(18) == 'r' &&
        password.charAt(13) == '3' &&
        password.charAt(19) == '4' &&
        password.charAt(21) == 'T' &&
        password.charAt(16) == 'H' &&
        password.charAt(27) == '6' &&
        password.charAt(30) == 'f' &&
        password.charAt(25) == 'l' &&
        password.charAt(22) == '3' &&
        password.charAt(28) == 'd' &&
        password.charAt(26) == 'f' &&
        password.charAt(31) == '4';
}
-:***- VaultDoor1.java Bot (17,0) (Java//l Abbrev)

```

I solved this with a one-liner in my shell:

```
% grep charAt VaultDoor1.java |sort -V |awk
{'print $3'} |tr -d "\"'\n"
LOL_FLAG_REDACTED_LOL
```

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1. Pingback: [picoCTF Writeups – DMFR SECURITY](https://dmfrsecurity.com/2021/11/01/picoctf-2019-vault-door-1-writeup/)

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